TASK 1 REPORT

Retail Sales Dataset — Data Cleaning and Preprocessing

Prepared by: Shruti Sharma

Tools: Python, Pandas, NumPy, Google Colab / Jupyter Notebook

Project Summary

This report documents the cleaning and preprocessing of a retail orders dataset before exploratory data analysis. The workflow included data inspection, missing-value review, duplicate removal, numeric validation, standardization, outlier detection, final checks, and CSV export.

Metric	Result
Raw dataset size	1,060 rows
Exact duplicate rows removed	60
Final dataset size	1,000 rows × 17 columns
Duplicate rows remaining	0
Final validated age range	5–64 years
Potential high unit-price outliers flagged	73
Extreme price records marked for verification	10

Outliers were flagged for review rather than automatically deleted. Unusually high values may represent legitimate transactions.

1. Objective

Convert raw retail sales data into a consistent, analysis-ready dataset, reducing errors in statistics, visualizations, and predictive modelling.

2. Dataset Overview

The dataset includes order identifiers, customer details, gender, age, city, product category, quantity, unit price, discounts, payment method, order date, customer ratings, and coupon information. The final output contains 1,000 records and 17 columns.

3. Tools Used

Python automated the workflow. Pandas supported inspection, transformation, missing-value handling, duplicate detection, and export. NumPy supported numerical operations. Matplotlib and Seaborn can be used for visual checks. The workflow was carried out in Google Colab.

4. Cleaning Methodology

Step	Work performed	Purpose
Inspect	Loaded file; checked rows, columns, sample records, and data types.	Understand data before editing.
Missingness	Counted missing values by column.	Identify fields needing attention.
Duplicates	Removed 60 exact duplicate rows.	Avoid inflating counts and totals.

Step	Work performed	Purpose
Validation	Checked numeric fields for invalid or suspicious values.	Improve consistency.
Age handling	Corrected 6 invalid ages and filled 93 missing ages using the median of valid ages.	Created a usable age field.
Standardization	Standardized column names/text fields and reviewed data types and dates.	Improved consistency and usability.
Outliers	Used IQR to flag unusually high prices and checked high quantities.	Identified values that may influence summaries.
Final validation/export	Rechecked counts and ranges, then exported CSV.	Confirm and preserve cleaned output.

5. Quantitative Results

Check	Observed result
Rows	1,060 initially; 1,000 after cleaning
Duplicates	60 removed; 0 remaining
Age	87 missing and 6 invalid initially; final range 5–64
Gender	43 missing values in the cleaning audit
City	29 missing values in the cleaning audit
Payment method	65 missing values in the cleaning audit
Unit price	Mean ■9,548.24; median ■1,098.65; maximum ■1,200,000
Outlier review	73 high-price flags; 10 extreme cases for verification; 4 high-quantity observations noted for review
Final structure	1,000 rows × 17 columns

6. Missing-Value Handling

Missing values were assessed by column. Age was handled after validating its range, with missing ages filled using the median of valid ages. Unknown gender, city, payment method, date, rating, or coupon values should not be guessed. Coupon Code had 880 missing values (about 88% of the cleaned dataset), so coupon-related analysis should be interpreted carefully.

7. Outlier Detection and Treatment

The Interquartile Range (IQR) method flagged 73 potentially high unit prices, including 10 extreme cases requiring verification. Four high-quantity observations were also noted for review. Outlier flags are prompts for investigation, not proof of errors. Check extreme transactions against source records where possible.

8. Final Validation

Validation	Outcome
Rows and columns	1,000 rows and 17 columns
Duplicate rows	0 remaining
Age values	No invalid tested ages remaining; range 5–64
Potential outliers	Flagged for review rather than indiscriminately removed
Export file	cleaned_retail_orders.csv

9. Key Learnings

• Start with a data-quality audit before making changes.
• Duplicate removal prevents repeated records from distorting counts and totals.

• Median imputation can be useful for numeric fields when based on valid values.
• Do not invent unknown customer or transaction details.
• Outlier detection identifies unusual observations; it does not automatically make them errors.
• Rechecking counts, ranges, duplicates, and data types helps verify cleaning.
• A cleaned CSV creates a reusable handoff for EDA and modelling.

10. Limitations and Recommendations

Some fields remain missing because their true values cannot be recovered from the available data. Unit price is strongly right-skewed: mean ■9,548.24 is much higher than median ■1,098.65, and the maximum is ■1,200,000. These values can influence averages and totals. Verify extreme records where possible and document exclusions. Distinguish gross sales (quantity × unit price) from net revenue after discounts, returns, taxes, or other adjustments.

11. Conclusion

The workflow reduced the dataset from 1,060 to 1,000 rows by removing 60 exact duplicates and produced a 1,000-row, 17-column cleaned dataset. Age values were validated, missing values were reviewed without guessing unknown details, and unusual price and quantity values were flagged for verification. The cleaned data provides a stronger foundation for Task 2 EDA and later modelling.

Deliverable: cleaned_retail_orders.csv